DATA SCIENCE FOR BIOLOGY · WORKSHOP II

Before It Reaches R

How DNA and RNA sequencing data is generated, and what that means for your analysis



Lindsay Droit
Washington University School of Medicine, St. Louis
Nairobi, Kenya · 9-13 November 2026

Your data remembers everything that happened to the sample.



How long it sat before freezing



Which kit, which lot, which day



Who handled it



Where it sat on the plate



Which run, which lane

In R, all of it shows up as variance. The question is always: is this variance biology, or history?

Today's roadmap

01	**The sequencing landscape**	What each layer measures and how
02	**From sample to molecule**	Collection, extraction, library prep, QC
☕	**Break**	15 min
03	**Where artifacts come from**	Batch effects, contamination, controls + Activity 1
04	**From instrument to data file**	Files, quality scores, depth, zeros
05	**Metadata and study design**	What to record, what to ask + Activity 2

01

The sequencing landscape

What each approach measures, and how the method shapes the data

PART 1

What's inside a sample

Every sample is a mix of cells, and every cell holds the same few kinds of molecules.

Today: DNA and RNA



One sample

contains many kinds of cells



Human



Bacteria



Fungi



Viruses





DNA

The cookbook

Every recipe the cell could ever make. Barely changes.



RNA

Today's photocopied recipe

What the cell is making right now.



Protein

The finished dish

What actually does the work.



Metabolites

Ingredients and leftovers

What the cell uses up and gives off.

PART 1

The DNA and RNA toolbox

Each method answers a different question about the same sample.

HUMAN BIOLOGY ASSAYS



Whole genome

DNA

What's in the whole cookbook?

Short- or long-read WGS



Targeted DNA

DNA

What's on a few pages?

Exome, panels, amplicons



Bulk RNA

RNA

What is the sample doing now?

RNA-seq

First select mRNA, or deplete rRNA



Single-cell

RNA

What is each cell doing?

Droplet or plate methods

First select mRNA, or deplete rRNA



The microbiome cuts across every layer: the same stool sample can be read for microbial DNA, RNA, and viruses.

PART 1

A short history of sequencing

Sequencing went from one piece of DNA at a time to billions at once, and the cost collapsed.

1977

FIRST GENERATION

Sanger sequencing

One piece at a time; built the first human genome.

2005

SECOND GENERATION

454

First next-gen sequencer: millions of pieces at once.

2007

Illumina (Solexa)

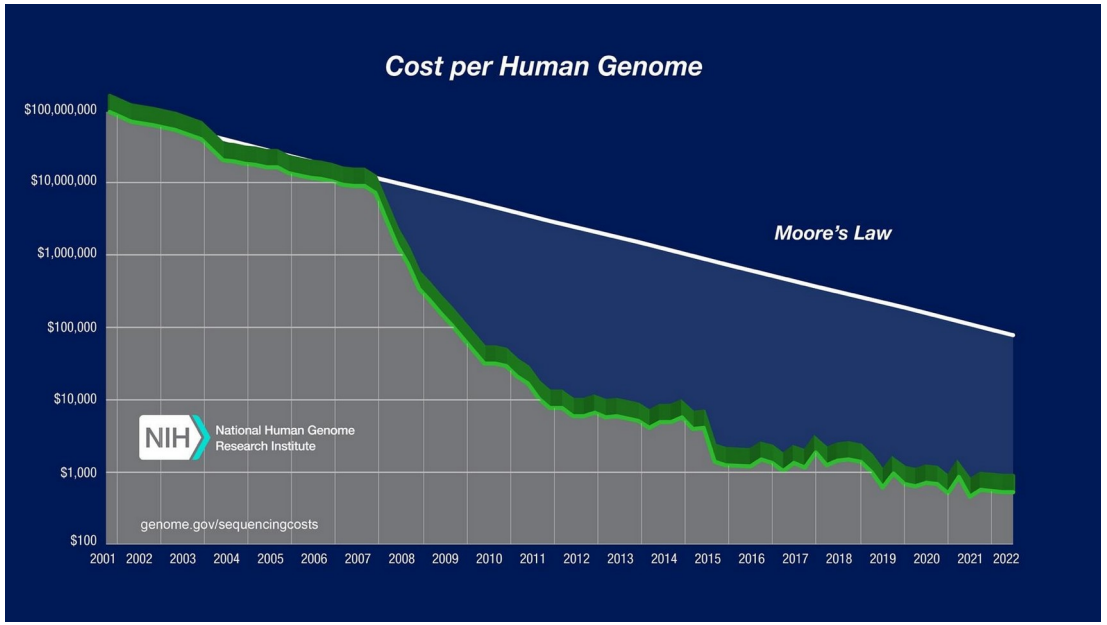
Short reads in huge numbers (MiSeq, NextSeq, NovaSeq).

2010s

THIRD GENERATION

PacBio, Oxford Nanopore

One long molecule at a time; Nanopore fits on a laptop.



Cost of one human genome, 2001-2022: from ~$100 million to a few hundred dollars.

Why it matters: cheap sequencing means bigger studies, more batches, and more room for batch effects.

Cost chart: NHGRI, genome.gov/sequencingcosts

What sequencing does, in one picture

Chop it up, label it, read it, then put it back together or count it.



1. Cut

Break DNA into millions of small pieces

2. Label

Add a barcode saying which sample each piece came from

3. Read

The machine reads each piece, letter by letter: A, C, G, T

4. Rebuild or count

A computer matches pieces to a reference, or counts them

Think of it like this: shred 100 different books, write the title on every scrap, read every scrap, then use the library catalogue to put each scrap back with its book.

PART 1

Short reads vs long reads

Short reads are cheap and accurate. Long reads show the big picture.

	Short read	Long read
Examples	Illumina, Element AVITI, MGI, Ultima	PacBio HiFi, Oxford Nanopore
Read length	100-300 letters	Thousands to 100,000+ letters
Best for	Counting, variants, most studies; lowest cost	Repeats, big rearrangements, full-length genes
Watch for	Can't span long repeats; PCR copies	More DNA needed; higher cost per sample

Single-end vs paired-end



Single-end One read per fragment

Paired-end Both ends read: R1 + R2

Why it matters: paired-end reads give more information for mapping, assembly, and telling similar sequences apart, but they produce more data and cost more.

Specs change yearly. Confirm with your sequencing provider.

PART 1

RNA-seq: choices that change your data

How the RNA library was made decides what you can and can't see.



Which RNA?

PolyA keeps messenger RNA and needs intact RNA. Ribosomal RNA depletion keeps everything else and tolerates damage. Depletion kits are species-matched: human or mouse for host RNA, bacterial for metatranscriptomes.



Which strand?

Stranded libraries record which DNA strand a gene came from. Tell your software.



Whole gene or just the end?

Full-length shows gene versions. 3' tag only counts genes, but costs less.



How many reads?

About 20-30 million per sample is typical for comparing gene levels.



How intact?

RIN scores RNA quality from 1 to 10. Low RIN + polyA = skewed data.

PART 1

Single-cell: four labels on every read

Each read says which sample, which cell, which original molecule, and which gene.

Sample label	Cell label	Molecule label	Gene
which library	which droplet	UMI: which original	the message

Think of it like this: a letter with a country, a street address, a tracking number, and the message inside.

What goes wrong

- **Doublets:** two cells in one droplet look like a new cell type
- **Empty droplets:** a 'cell' that is really just background
- **Loose RNA:** free-floating RNA ends up in every droplet
- **Stress genes:** cutting up tissue stresses the cells

02

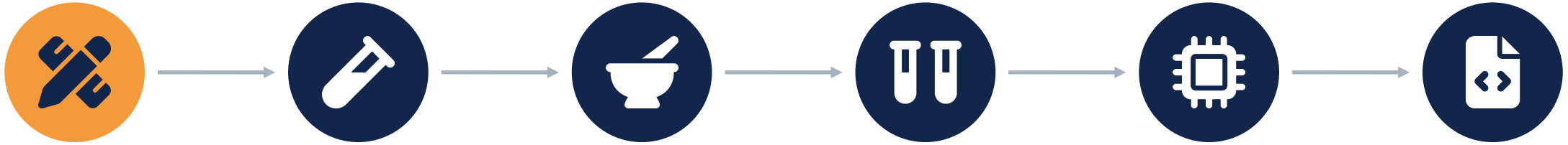
From sample to molecule

Collection, extraction, library preparation, and the QC in between

PART 2

The data generation pipeline

Every sample takes the same six steps from person to file.



Design

Plan groups, numbers, and batches

Collect

Take the sample, freeze it, ship it

Extract

Break cells open, pull out DNA / RNA

Prepare

Label and package molecules for the machine

Measure

The sequencer reads it

Data

Files and tables you receive

Rule of thumb: analysis can fix some bench problems, but only if they were written down and don't line up with the groups you compare. The earlier the mistake, the harder it is to fix.

PART 2

Collection and storage

Samples keep changing after collection. Freeze them fast, and the same way every time.



Field site, Marsabit, Kenya: the nearest -80 °C freezer can be hours away.

Think of it like this: milk left on the counter. The longer it sits, the more it changes.

Time to freeze

Write it down. Minutes matter for RNA.

Temperature

-80 °C as fast as possible. Avoid thawing and refreezing.

Preservative

Pick one (or none) and use it for every sample.

Shipping

Dry ice with a temperature log. Same route for all sites.

Amount

Collect extra. Stool: ~20 mg for 16S, ≥200 mg for virome.

PART 2

Handling shows up in the data

If samples were handled differently, the data will show it, and it can look like biology.

What happened	What you see in the data	Affects
RNA damaged (low RIN)	Skewed coverage, false gene differences	RNA
Stool left warm	Some bacteria keep growing	Microbiome
Thawed and refrozen	Broken DNA and RNA; RNA viruses lost	All
Very small sample	Stray DNA makes up more of the reads	Microbiome, low-input RNA

The fix: write these down for every sample, so you can check for them later.

One sample, DNA and RNA

Each kind of molecule needs different handling, so split the sample on day one.



Parent sample
ID: P017-V1

Host DNA

P017-V1-A

Freeze -80 °C; forgiving

Microbiome DNA

P017-V1-B

Freeze -80 °C, no additive

RNA

P017-V1-C

Stabilizer or flash freeze

Virome

P017-V1-D

Large portion, never refreeze

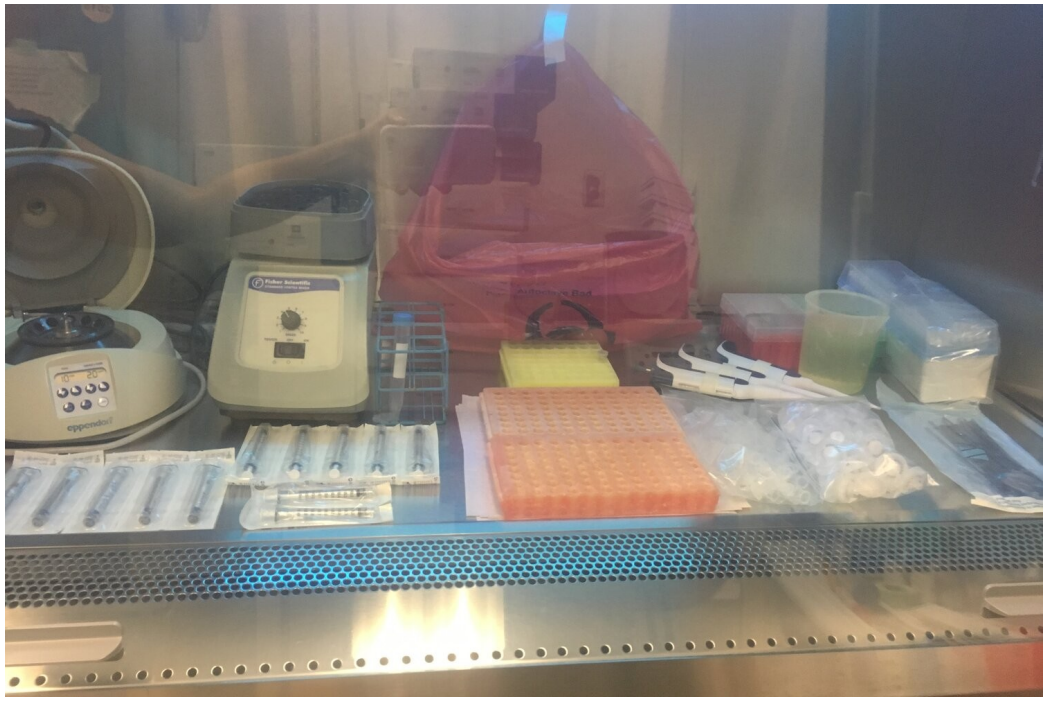
Plan portions on day one

- Split at collection, not later
- Handle each portion for its own purpose
- Every portion keeps the parent ID
- Same steps at every site
- Keep a spare for follow-up

PART 2

Extraction

If a cell doesn't break open, its DNA never makes it into your data.



Our biosafety cabinet set up for stool processing.

Tough cells need force. Without bead beating, thick-walled bacteria are under-counted.

Kits differ. Same sample, different kit, different answer.

Some samples fight back. Stool contains chemicals that block PCR; diluting sometimes helps.

Write it down. Kit, lot number, operator, and day.

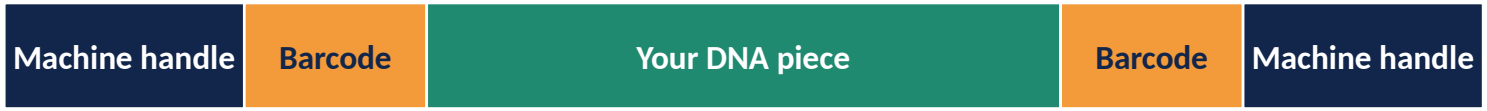
Think of it like this: cracking nuts. Some open with your fingers; others need a hammer. Bead beating is the hammer.

PART 2

Shotgun library preparation

Cut the DNA into pieces and attach handles and barcodes. Amplicon 16S/ITS uses its own PCR-based route.

Fragment	Tidy ends	Add adapters	Wash	Amplify	Check + pool
or tagment	ready to attach	if not tagmented	magnetic beads	make copies, if required	then mix samples

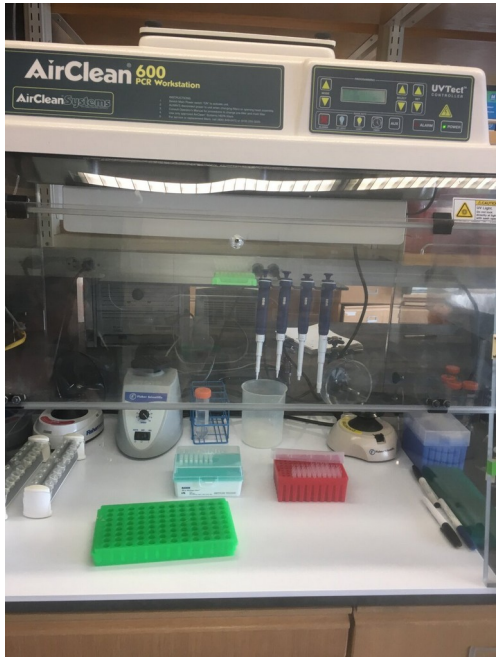


One finished library piece

A barcode is a short, known stretch of DNA, unique to each sample and read along with the fragment. You need one because a single run makes far more reads than any one sample needs, so samples are pooled to share the cost, and the barcode is the only way to sort them out again. Like a return address on an envelope.

Wrong labels: a few reads get another sample's barcode. Two barcodes per sample catch this.

Too many copies: small samples need more PCR, so the same piece gets read many times.



Our PCR workstation

Three ways to sequence a microbiome

Read one gene, read everything, or filter out all but the viruses.

16S / ITS (amplicon)

A roll call: read one name-tag gene from everyone

- Who is there (usually genus)
- ~20 mg stool
- Cheap: thousands of samples

Watch for: Genus-level only; primer choice biases who you see

Shotgun metagenomics

Record the whole room: read all the DNA

- Species, strains, and genes
- Human DNA can dominate
- Costs much more per sample

Watch for: Rare microbes and viruses get drowned out

Virome

Clear the room of everything bigger than a virus

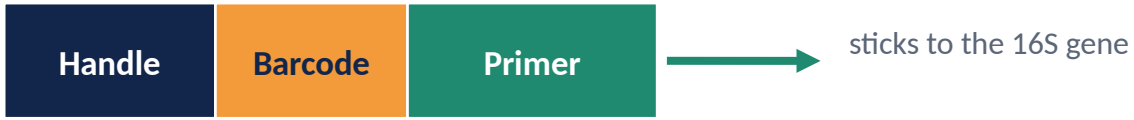
- DNA and RNA viruses
- ≥200 mg stool
- Sees what shotgun misses

Watch for: Tiny amounts: contamination risk

Amplicon sequencing, step by step

One PCR copies the name-tag gene and adds the sample barcode at the same time.

The barcoded primer



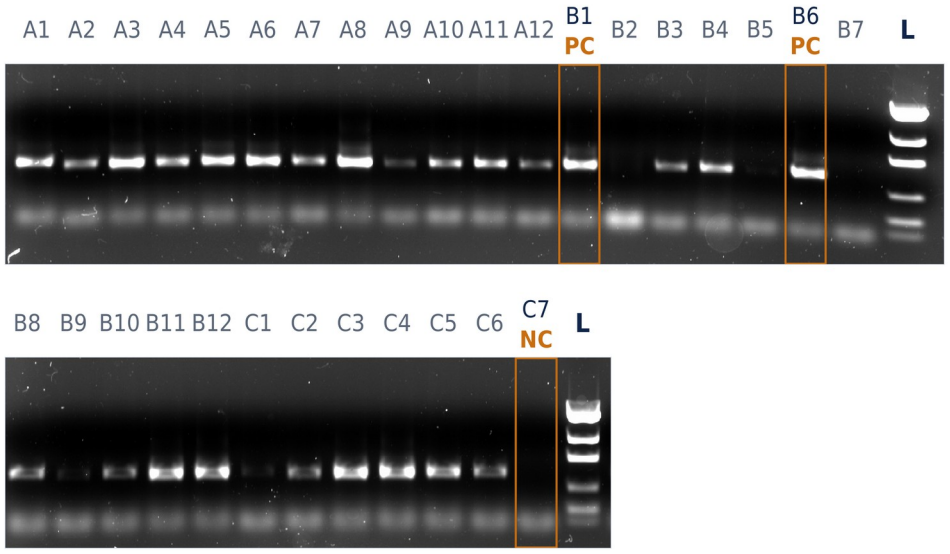
Run each sample 3 times, plus a plate of water blanks

Check on a gel: one clear band means PCR worked

Up to 40% can fail, e.g. samples from people on antibiotics

Blocked PCR? Diluting 1:10 often rescues it

Pool the same volume of every sample, blanks included



PC	positive control, a sample with a known answer
NC	no-template control, this lane has to stay empty
L	size ladder. The 16S V4 product runs at about 390 bp

Primer design after Caporaso et al., PNAS 2011 and ISME J 2012 (Earth Microbiome Project)

Virome: enrich first, then sequence

Remove everything bigger than a virus, destroy loose DNA, then read what's left.

1. **Chip ~200 mg** stool into buffer
2. **Vortex** break it up, don't burst cells
3. **Spin** 7,000 g, 10 min
4. **Filter** 0.45 µm removes bacteria
5. **Enzymes** destroy loose DNA
6. **Extract** DNA and RNA
7. **Convert** RNA → DNA
8. **Copy + library** ready to sequence

Think of it like this: panning for gold. Wash away the dirt and keep the tiny valuable bits.

Why bother?

In regular shotgun data, viruses are a tiny fraction, and RNA viruses are invisible.

The price

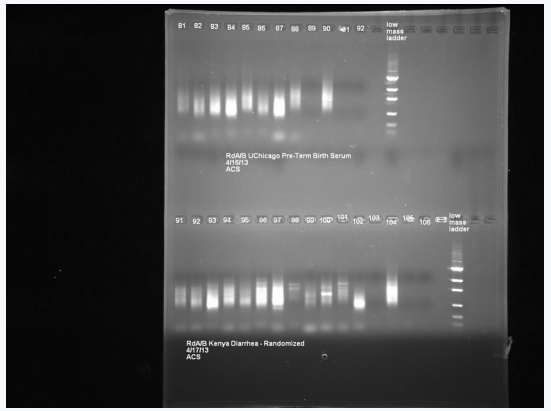
Tiny amounts mean stray DNA matters more. Many reads match nothing in any database.

PART 2

Lab checks before sequencing

Three checks at three different points. Ask which one a number came from.

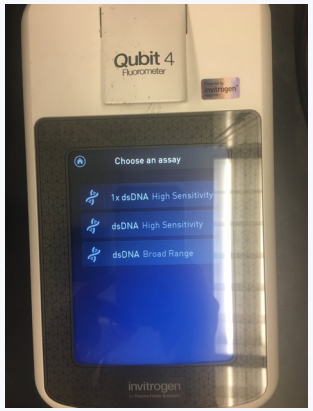
Extract → RdA/B or PCR → **gel** → clean up → **Qubit** → library prep → **Bioanalyzer** → sequence



after Round B, on 10 µL

Did it work?

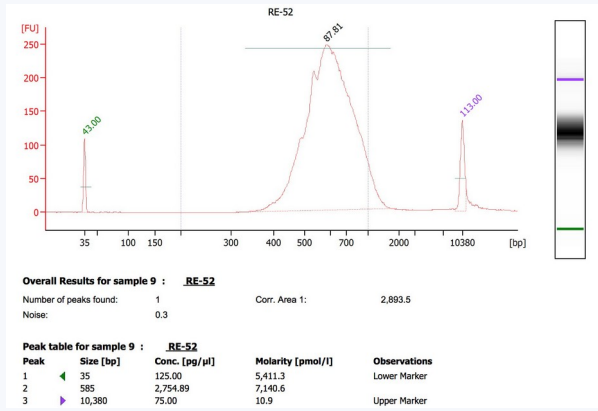
A band or smear means amplification worked, and the blanks should have stayed empty. This is the first place a failed sample shows itself.



after cleanup, before library prep

How much?

Qubit measures what you actually recovered, and sets how much goes into the library, typically 20 to 100 ng. Short here means short on reads later.



on the finished library

What size?

Bioanalyzer or TapeStation shows the size spread. One broad hill is good, a sharp small spike is leftover adapter. For RNA, run the input RNA too, for a RIN score.

Watch how an Illumina sequencer works

One short video. Watch for these three things; each one shows up in your data.



Illumina: sequencing by synthesis

youtube.com/watch?v=fCd6B5HRaZ8

Click to open in your browser

Watch for

1. **One cycle, one letter.**
   That's why reads are short, and why you see "2 × 150".

2. **The signal fades each cycle.**
   That's why quality drops toward the end of a read.

3. **Many samples share one flow cell.**
   That's where barcode mix-ups come from.

Break

15 minutes. When we come back: where artifacts come from, and your first activity.

03

Where artifacts come from

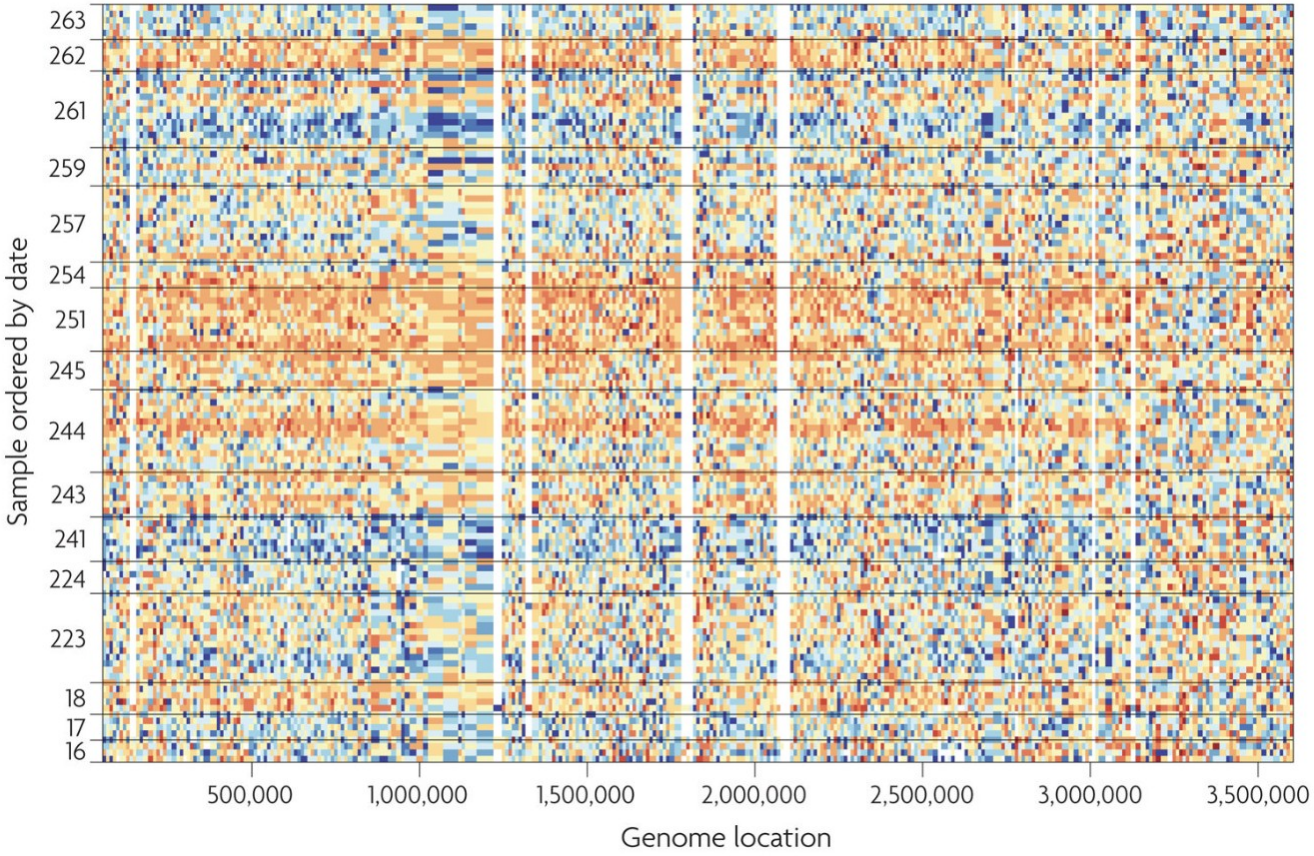
Batch effects, contamination, controls, and your first activity

PART 3

Batch effects

Samples processed together share quirks. That's fine, unless the quirks line up with your groups.

- **Well position on the plate**
- **Plate**
- **Extraction batch** day, kit lot, operator
- **Library prep batch**
- **Sequencing run** flow cell and lane
- **Sequencing centre** or collection site



Each row is one sample. Colour is coverage: blue low, orange high. Black lines separate processing days.

Whole days run warm together. Same lab, same machine, different day.

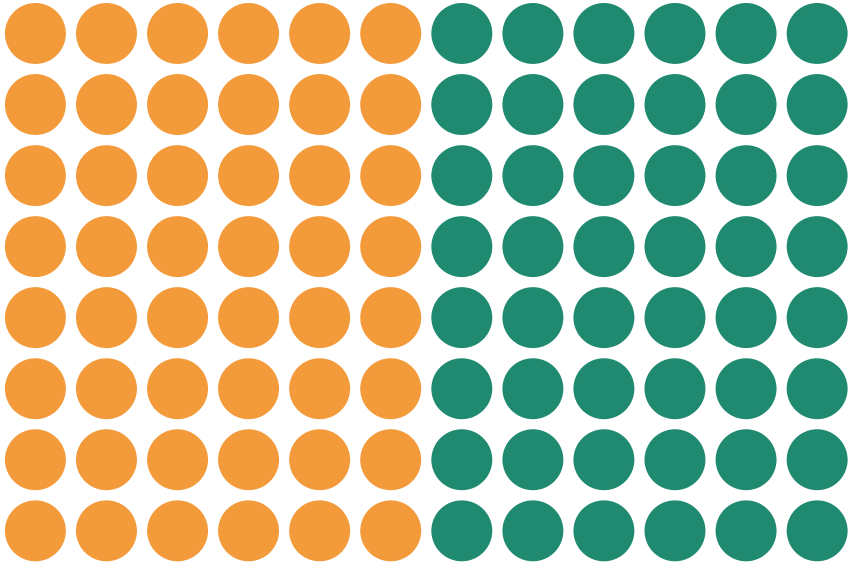
Leek et al., Nat Rev Genet 2010, Fig. 2, reproduced with permission. 1000 Genomes, 3.5 Mb of chromosome 16.

PART 3

Plate layout: confounded vs balanced

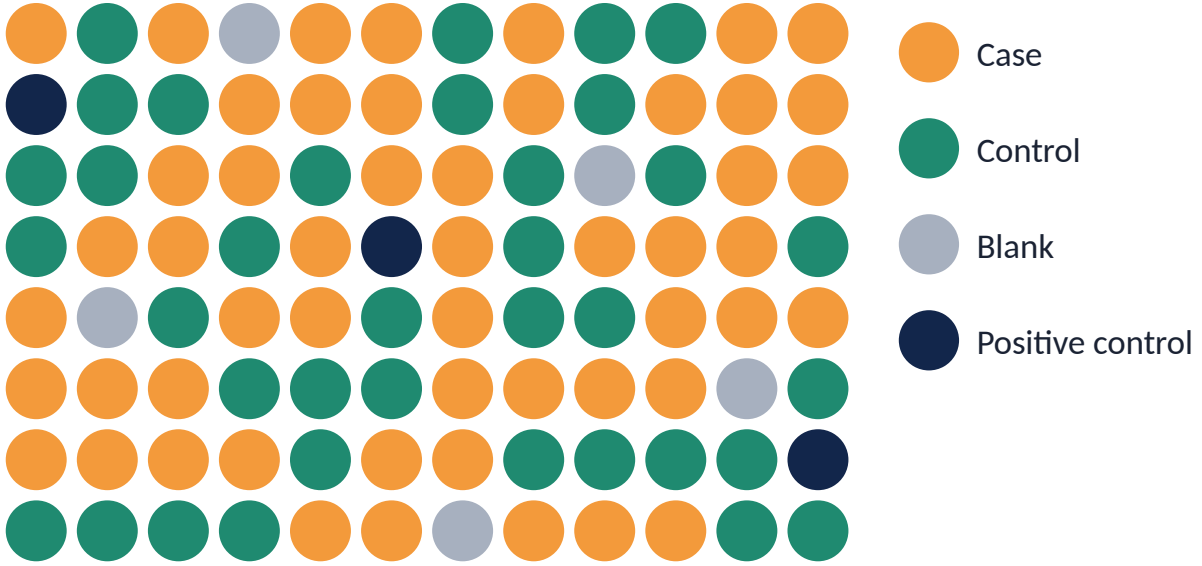
Mix your groups across the plate, so a position on the plate can't pass for biology.

Confounded



Filled in order of arrival: cases in columns 1-6, controls in 7-12, no blanks.

Balanced



Randomized; blanks and positive controls scattered across the plate.

Multiple plates? Balance every plate for group, site and timepoint. Keep all timepoints from one participant on the same plate.

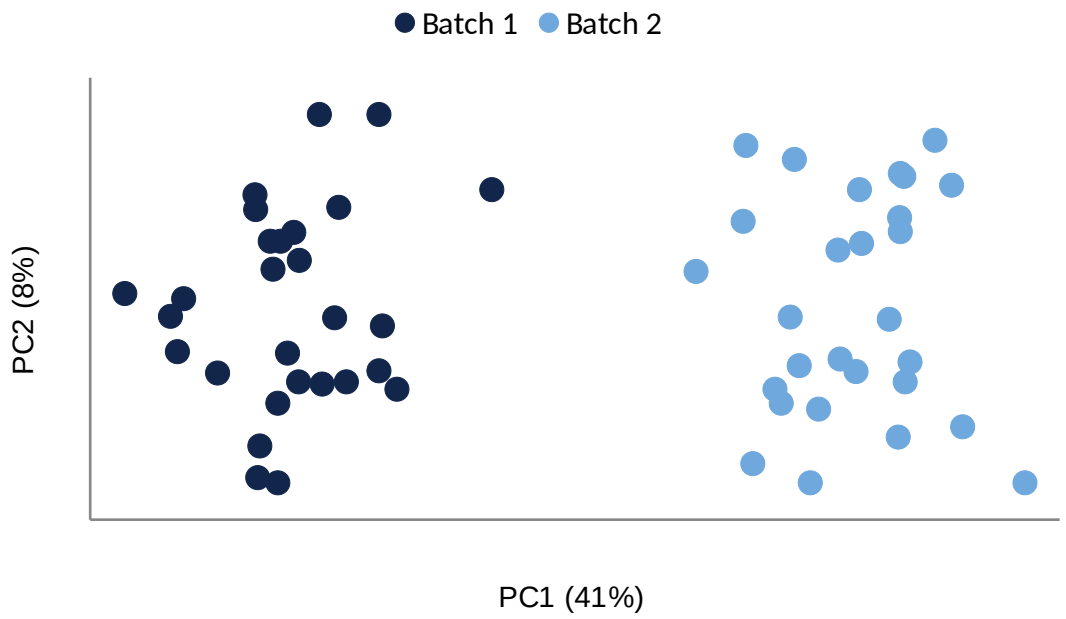
PART 3

What a batch effect looks like in a PCA

The same samples twice. Only the colour changes.

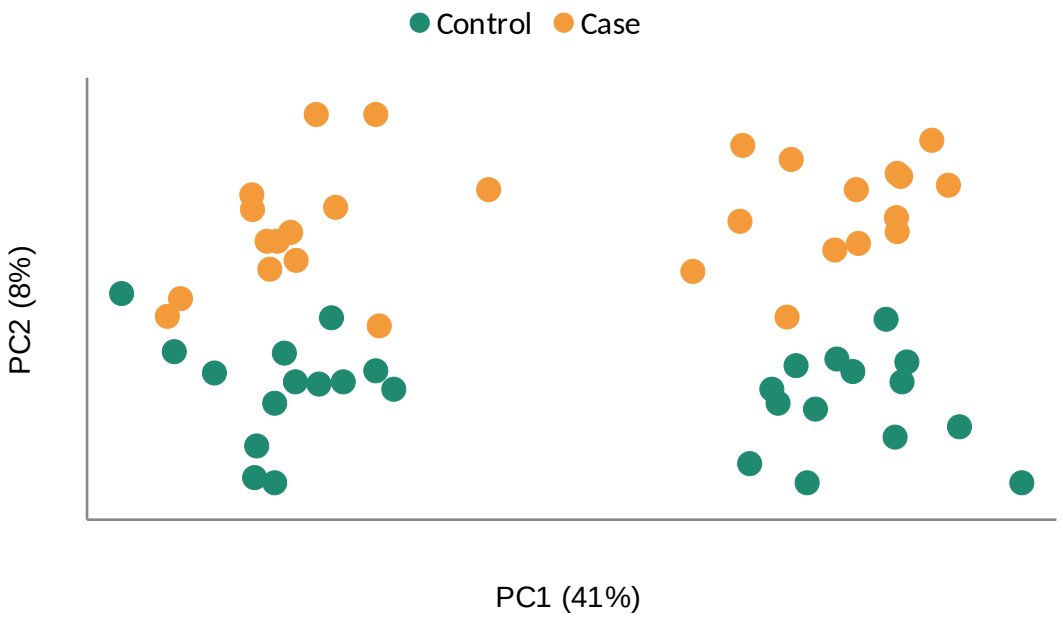
Coloured by extraction batch

Two tight clusters, split left to right along PC1.



Coloured by biological group

Both colours show up in both clusters.



Habit: colour your first PCA by every technical variable in the metadata before you colour it by biology.

PART 3

Two questions decide what analysis can rescue

A map for later in the week. You will come back to this when you start correcting for batch in R.

	Written down the variable is in your metadata	**Not written down** nobody recorded it
Balanced both groups appear in every batch	You can fix it	You see it, can't name it
Confounded each batch is mostly one group	You know you're stuck	You never find out

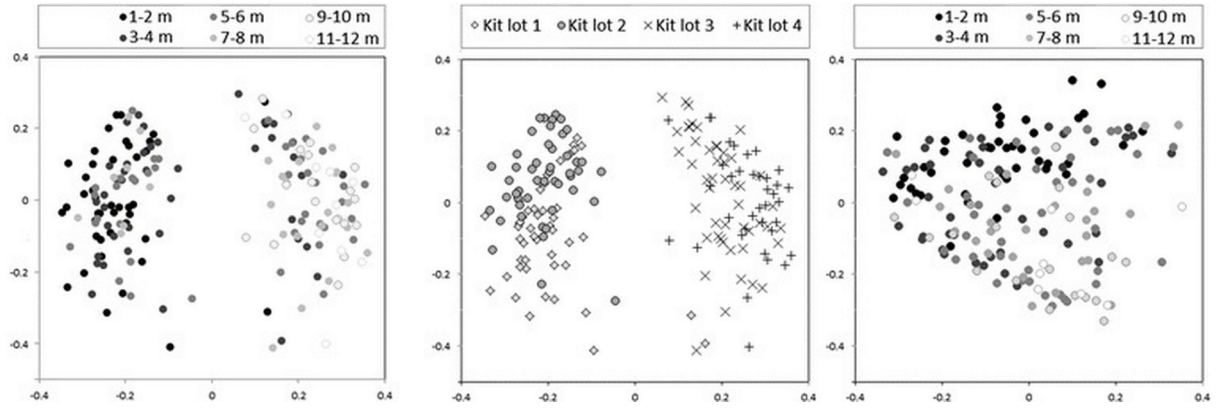
To tell two things apart, you need at least one sample that has one of them without the other.

PART 3

Contamination

Every kit carries a little stray DNA. With tiny samples, it can be most of what you read.

A real example: infant nose and throat swabs, sampled through the first year



a) coloured by age b) coloured by extraction kit lot c) contaminants removed

The babies appeared to cluster by age. They were clustering by kit lot.

Take the contaminant OTUs out and the age pattern is gone.

Salter et al., BMC Biol 2014, Fig. 4a-c, cropped. CC BY 4.0.

Think of it like this: a whisper is easy to hear in a quiet library and lost in a busy market. Tiny samples are the quiet library.

Plenty of DNA: safe

- Stool
- Soil
- Wastewater

Very little DNA: at risk

- Breastmilk
- Spinal fluid
- Lung
- Blood
- Skin
- Tissue

Sources: kits and reagents, neighbouring wells, wrong barcodes, people.

Low-biomass standards: Nature Microbiology consensus guidelines, 2025

PART 3

Controls

Controls show what sneaks in, whether the method works, and how much noise there is.

Negative control

A blank tube that goes through every step.

Shows what sneaks in.

Examples: Extraction blank, PCR water blank

Positive control

A sample where you already know the answer.

Shows the method works.

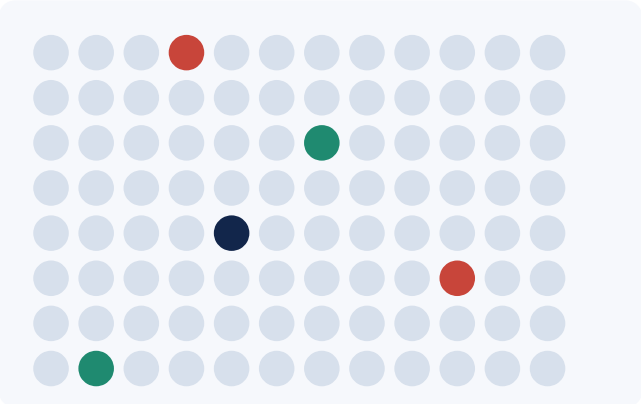
Examples: Mock community, Reference RNA, Reference genome

Replicates & spike-ins

Run a sample twice, or add a known amount to every sample.

Shows noise; gives a ruler.

Examples: Technical replicates, ERCC spike-ins, spike-in cells



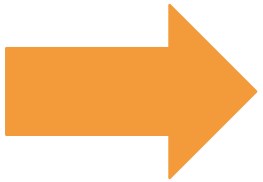
One plate, controls scattered

Lab layout: one-way traffic

Copied DNA is the biggest contamination risk in a lab, so it only ever moves one way.

Clean room (before copying)

- Opening original samples
- Extracting DNA and RNA
- Setting up PCR



one way only

Main lab (after copying)

- Running PCR
- Gels and library building
- Checking and pooling

Think of it like this: a kitchen that keeps raw chicken away from the salad. Nothing from the main lab goes back: not tubes, pipettes, lab coats, or notebooks.

GROUP ACTIVITY · 15 MIN

Activity 1: Spot the problem

Three real-world mistakes. For each one: what went wrong, and can it be fixed?

Case 1: Two clusters

Your PCA shows two tight clusters that match the two extraction days perfectly. All HIV+ participants were extracted on day 1; all HIV- on day 2.

Case 2: Busy blanks

Your extraction blank has 25,000 reads, mostly Ralstonia and Bradyrhizobium. Your breastmilk samples average 8,000 reads, dominated by the same genera.

Case 3: Site effect

Site A RNA has RIN 8-9. Site B shipped on wet ice for 4 days: RIN 3-5. You find 1,200 genes 'differentially expressed' between sites.

For each case: 1. What went wrong? 2. Can analysis in R fix it? 3. What should have been done?

DEBRIEF

Activity 1: What we'd say

If it was written down and balanced, it can be fixed. If it lines up with the groups, it can't.

	What went wrong	Can R fix it?	Prevention
Case 1	Processing day lined up exactly with HIV status	No. Nothing can separate the day from the disease.	Mix HIV+ and HIV- samples on both days
Case 2	Stray DNA from the kit swamped tiny samples	Partly: remove what's in the blanks (e.g. decontam). Some samples may be unusable.	More blanks, more sample, one kit lot
Case 3	Shipping damaged Site B's RNA, creating false gene differences	Mostly no. RIN never overlaps between sites, so damage and site can't be separated.	Same shipping for every site; check RIN when samples arrive

Pattern: the design decides what analysis can rescue. Recorded + balanced = fixable. Unrecorded or confounded = not.

04

From instrument to data file

What lands on your laptop and what the numbers mean

PART 4

What lands on your laptop

First a big file of reads, then a table of numbers. Know what happened in between.

Layer	Raw file	Processed matrix	Typical R home
Genome	FASTQ → BAM / CRAM	VCF (variants)	`VariantAnnotation, GenomicRanges`
Bulk RNA	FASTQ	Gene × sample counts	`SummarizedExperiment, DESeq2, edgeR`
Single-cell	FASTQ	Gene × cell sparse matrix	`Seurat, SingleCellExperiment`
Amplicon	FASTQ	ASV table + taxonomy	`phyloseq, dada2`
Shotgun metagenome	FASTQ	Taxon / gene abundance table	`phyloseq, TreeSummarizedExperiment`
Virome	FASTQ	Contigs + viral abundance table	`phyloseq`
Nanopore	POD5 → FASTQ → BAM	As for the layer above	`Rsamtools, GenomicAlignments`
Combined	All of the above	Matched matrices + sample map	`MultiAssayExperiment, MOFA2, mixOmics`

Always ask: raw or already normalized? Which pipeline, which version, which parameters? For Nanopore, also which basecaller model.

PART 4

Anatomy of a FASTQ read

Every read is four lines: a name, the letters, a spacer, and a confidence score for each letter.

```
@LH00123:45:22FKW3LT3:3:1101:12345:1016 1:N:0:ACGTAC
GTGCCAGCAGCCGCGGTAATACGTAGGGTGCAAGCGTTAATCGG
+
FFFFFFFF:FFFFFFFFFF,FFFFFFF:FFFF:F,FFFFF:,FF
```

- **Line 1: name:** which machine, run, and lane. Your batch information hides here.
- **Line 2: letters:** the DNA sequence itself.
- **Line 3: spacer:** just a '+' sign.
- **Line 4: confidence:** one symbol per letter, coding how sure the machine is.

Quality score (Phred)

How sure the machine is about each letter

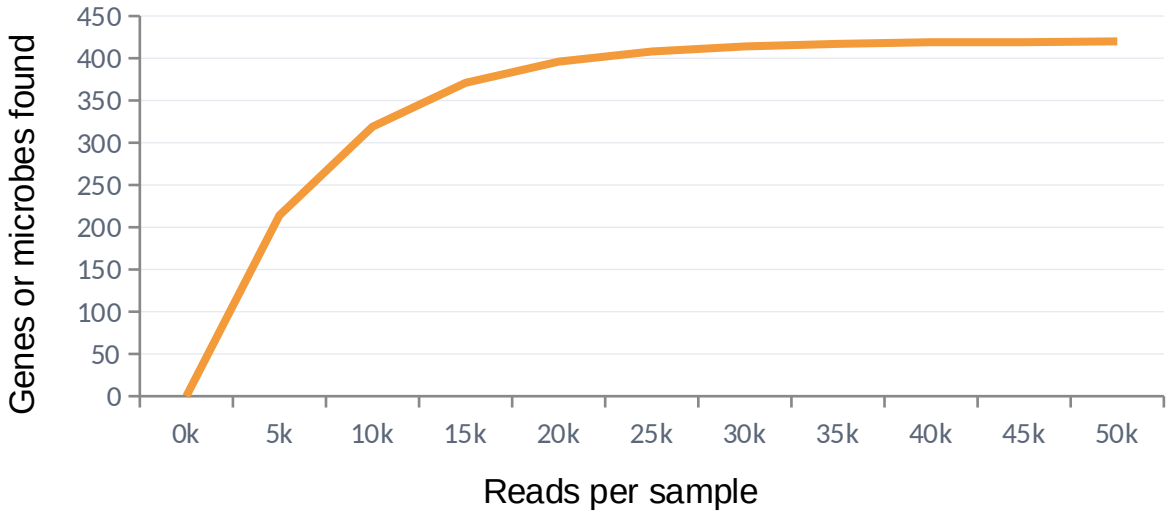
Q20	1 in 100 letters wrong
Q30	1 in 1,000 letters wrong
Q40	1 in 10,000 letters wrong

PART 4

How deep is deep enough?

More reads find more, until each extra read adds very little.

More reads, more found (simulated)



Depth depends on the sample, not just the assay

Human reads ranged 8.9% to 48.5% between women in one cervicovaginal study.

Run a few samples, then set the depth.

Typical starting points

16S amplicon	10,000-50,000 reads
Shotgun metagenome	5-20 million+
Bulk RNA-seq	20-30 million
Single-cell RNA	20,000+ per cell
Human genome	30× coverage

Depth How many reads each sample got.

Coverage How well those reads cover the sequence or features you care about.

Examples: 30M RNA-seq reads (depth) · 30× genome (coverage)

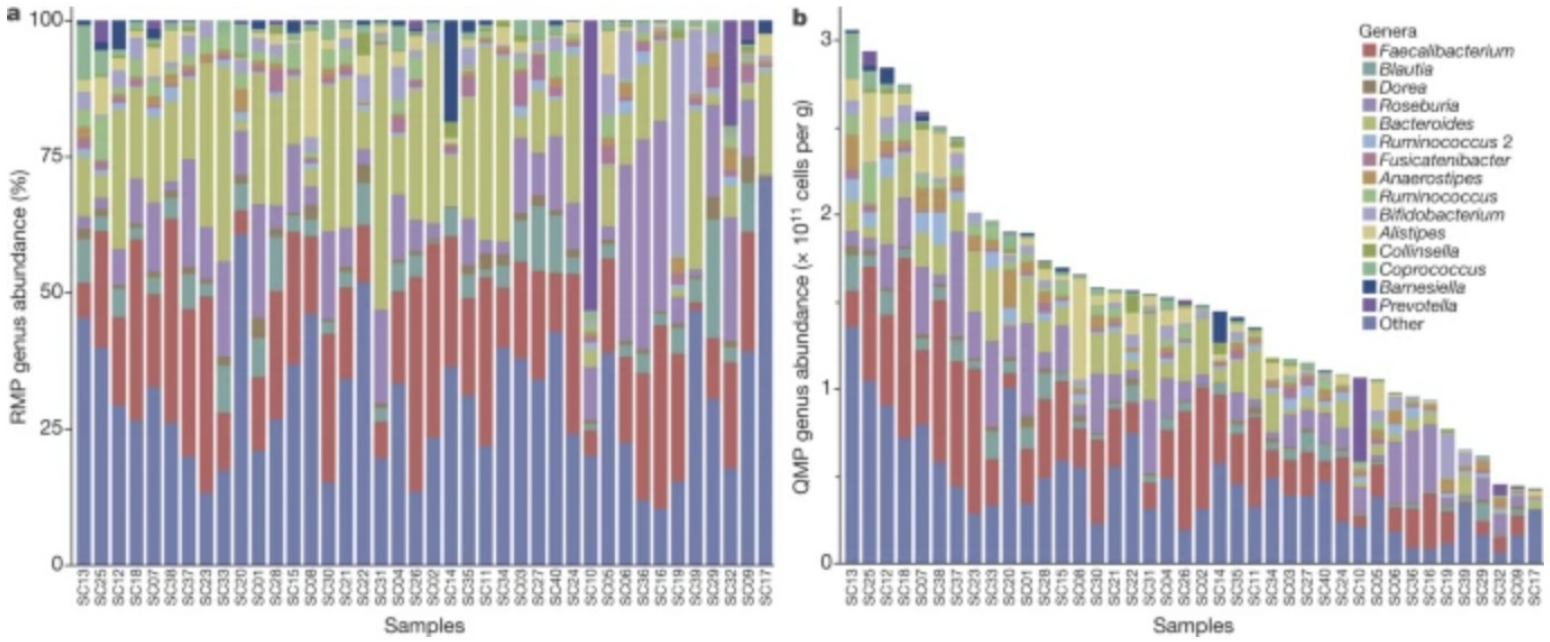
Bommana et al., Microbiol Spectr 2022

PART 4

Proportions, not amounts

The sequencer tells you the slices of the pie, not the size of the pie.

Forty real stool samples, read both ways



a Every bar is exactly 100%. That is all the sequencer gives you.

b The same samples in the same order, as real cell counts. Totals run from about 3.0 down to 0.5, a sixfold spread across these forty that panel a hides completely.

So...

- If one goes up, the others look like they went down
- Need real amounts? The lab must measure them (spike-ins, qPCR)

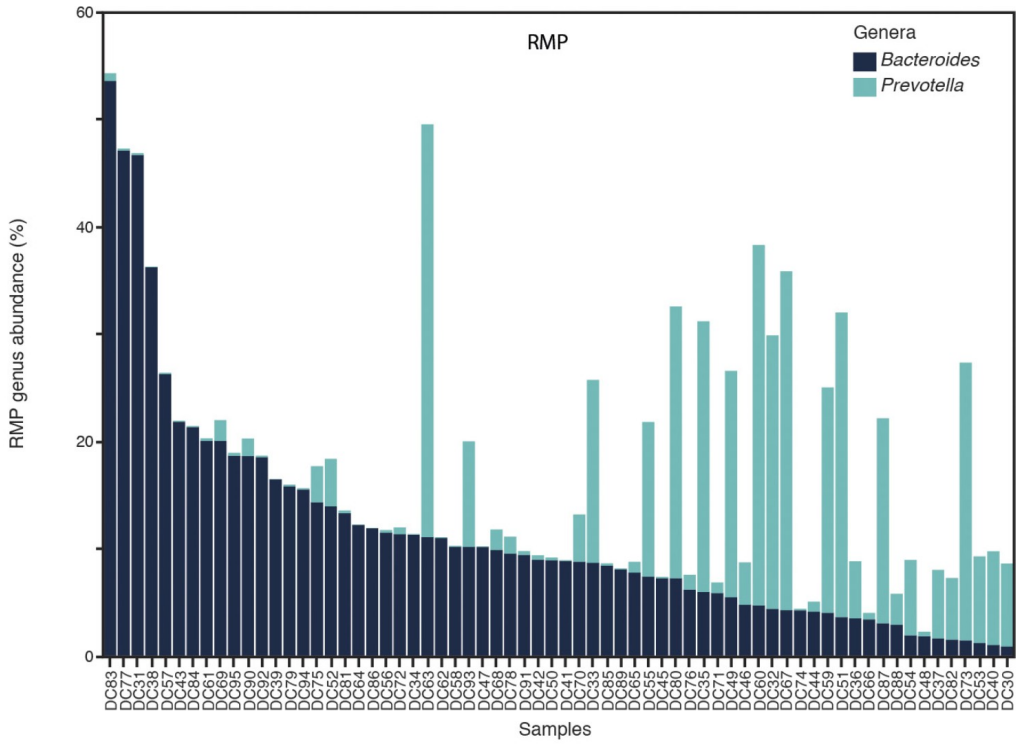
Vandeputte et al., Nature 2017, Fig. 2, reproduced with permission. *Gloor et al., Front Microbiol 2017*

PART 4

When proportions changed the answer

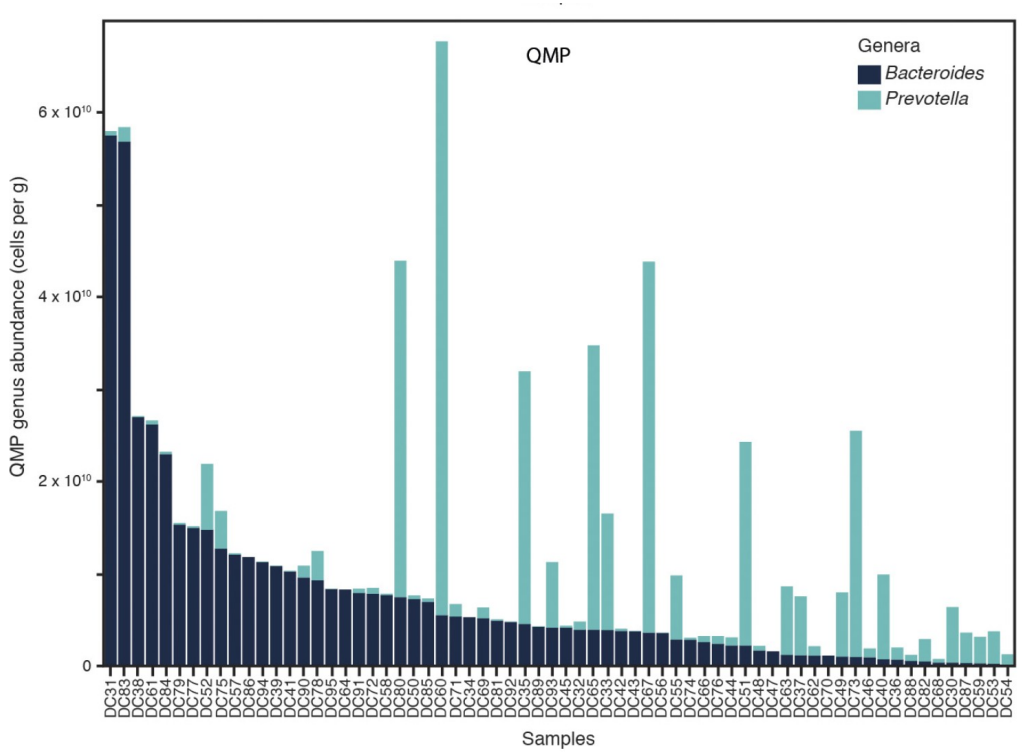
Same people, same sequencing. Two ways of looking at it.

As percentages (RMP)



Bacteroides vs Prevotella: ρ = −0.59, FDR 2×10^{-4}

As actual cell counts (QMP)



Same two genera: ρ = −0.33, FDR = 1, not significant

Percentages have to add up to 100. When one organism is low, everything else looks high, whether or not anything happened to it.

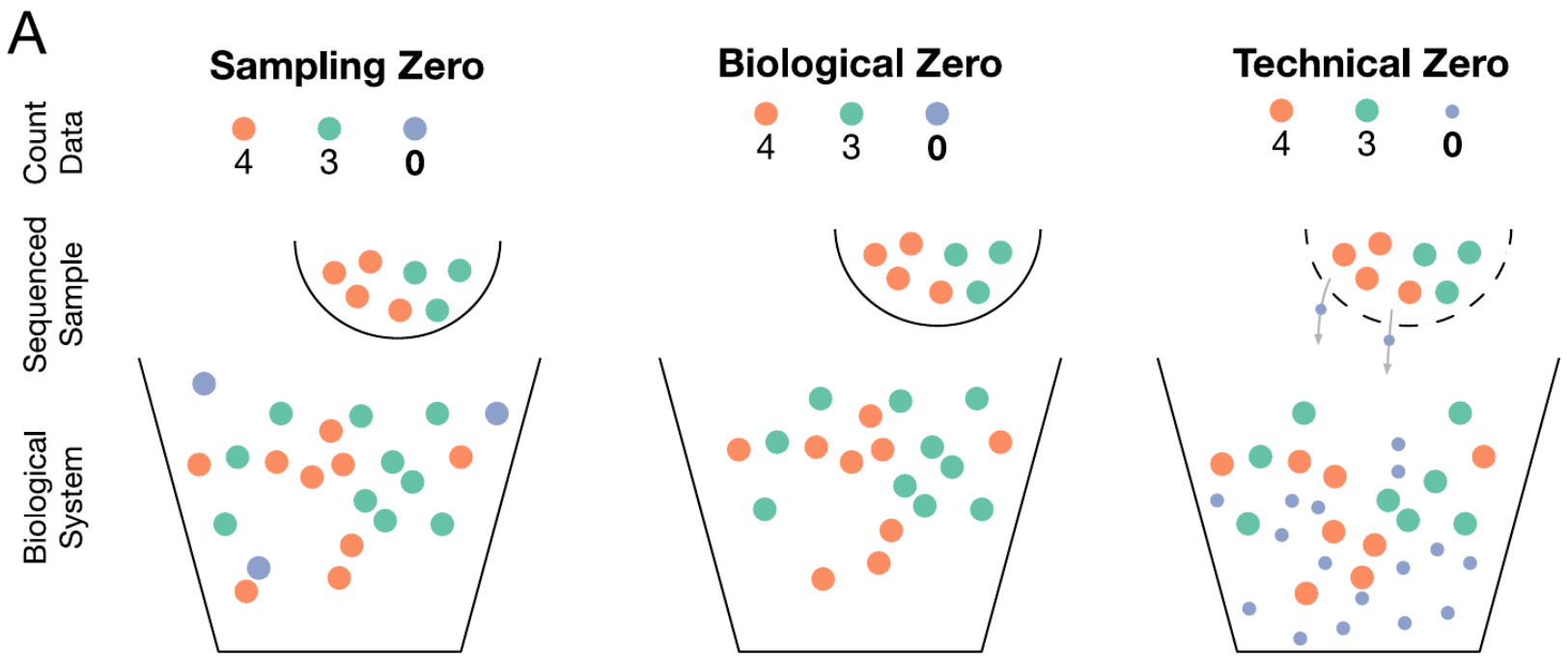
Vandeputte et al., Nature 2017, Extended Data Fig. 7, reproduced with permission. Healthy controls, n = 66.

PART 4

Zeros: absent, or just missed?

A zero or a blank has more than one cause. Ask which before you filter or fill it in.

Three ways a count can be zero. Read each column from the bottom up.



A FOURTH, NOT IN THIS FIGURE

Not in the database

Reads were there but matched nothing we know. The same failure as a primer that never matches an organism: it was in the sample, the method had no way to name it.

Think of it like this: not seeing a bird in ten minutes doesn't mean there are no birds.

What it changes: the zero model you choose changes which sequences come out significant.

Silverman et al., Comput Struct Biotechnol J 2020, Fig. 2A, reproduced under CC BY-NC-ND 4.0. "Naught all zeros in sequence count data are the same"

05

Metadata and study design

What to record, what to ask, and your second activity

PART 5

Metadata is data

If nobody wrote it down, you can't check for it later.

Sample

- Sample ID + participant ID
- Sample type, body site
- Collection date and time
- Site / clinic
- Time to freeze, preservative

Lab

- Extraction date + batch
- Kit and lot numbers
- Operator
- Plate ID + well position
- Input amount, QC values

Run

- Run ID, flow cell, lane
- Instrument
- Reads per sample
- Pipeline + version
- Control sample flags

One row per sample, one sample ID everywhere. Follow a standard (e.g. MIxS) from day one.

PART 5

Spreadsheet hygiene

Spreadsheets quietly change your data. A few habits prevent most of it.

What you typed → what Excel saved

What you typed	What Excel saved
SEPT2	2-Sep
MARCH1	1-Mar
1-10	10-Jan
007	7

HGNC renamed these symbols (SEPTIN2, MARCHF1), announced 2020, partly because of this.

Rules that prevent most problems

- One ID format, letters first: P017-V1, not 17 or 1-10
- Dates as YYYY-MM-DD, in a text column
- One variable per column, one sample per row
- No merged cells, colours, or notes-as-formatting
- Blank means blank: write NA, not 0
- Save a CSV copy; never edit the raw file

Ziemann et al., Genome Biol 2016; Bruford et al., Nat Genet 2020

PART 5

Ten questions to ask before you analyse

Send these to the lab before you start. They catch most problems early.

1. How were samples collected, stored, and shipped?
2. How were samples split into batches?
3. Which kits and lots? Did anything change?
4. Where are the controls, and their data?
5. Is there a plate map for every plate?
6. What quality numbers exist per sample?
7. Which machine, run, and lane?
8. Which software and version made the table?
9. Are the numbers raw or already adjusted?
10. Were any samples dropped or re-run? Why?

GROUP ACTIVITY · 15 MIN

Activity 2: Design the study

Plan it so that nothing about handling lines up with the things you want to compare.

The scenario

120 adults at two clinics (Nairobi and a rural site 6 hours away). Stool and blood at enrolment and 6 months.

You want: gut microbiome (shotgun metagenomics), gut virome, and blood transcriptome (RNA-seq).

Each extraction plate holds 96 wells.

Collection

How will each sample be collected, split, preserved, and shipped from both sites?

Batches

How many samples in total? How many plates? What goes on each plate?

Controls

Which controls, where, and how many per plate?

Metadata

Name 10 columns your metadata sheet must have.

DEBRIEF

Activity 2: One good answer

There's no single right answer. These are the choices that matter most.

Collection

- Split into aliquots at collection, one parent ID
- Same steps at both sites; if the rural site can't freeze fast, both use the same stabilizer
- Stool: record time to freeze, ship on dry ice with temp log
- Blood for RNA: PAXgene (or similar) tube at the draw

Batches

- 120 × 2 timepoints = 240 samples per sample type
- 84 samples + 12 controls per plate → 3 plates per sample type, 9 plates in total
- Balance site and timepoint on every plate
- Both timepoints of a participant on the same plate
- Trade-off: enrolment samples wait ~6 months in the freezer

Controls (per plate)

- Extraction blanks + library blanks
- Mock community / reference sample
- A bridging sample repeated on every plate

Metadata

- Sample, participant, site, timepoint, collection time
- Time to freeze, preservative, shipment
- Extraction batch, kit lot, plate, well, operator
- Run, lane, reads, pipeline version

Five things to take into the rest of the week

1. **Your data remembers.** Every step leaves a fingerprint. Ask 'biology, or history?'
2. **Design beats correction.** Balanced + recorded batches are fixable. Confounded ones are not.
3. **Controls are data.** Ask for them, sequence them, analyse them.
4. **Know what your numbers mean.** Counts are proportions; zeros have different causes.
5. **Talk to the lab.** Ten questions before you analyse will save you weeks.

Thank you

Questions?

Acknowledgements

Africa Health Research Institute (AHRI)
Afrah Khairallah · Tebogo Taueatsoala

Washington University School of Medicine in St. Louis · Handley Lab
Scott Handley · Luis Chica · Megan Johnson · Doha Abid · Leran Wang

The Ragon Institute · Kwon Lab
Doug Kwon · Joseph Elsherbini · Eyerusalem Tesfaye · Kwon Lab members

Harvard University Center for AIDS Research
Alberto Ottoboni · Meral Pasha

Workshop organizers and sponsors



Lindsay Droit · Washington University School of Medicine

Glossary

Plain-language definitions for the terms used today. Print it or photograph it.

Amplicon	A copy of one specific piece of DNA, made by PCR
Barcode / index	A short DNA tag that says which sample a read came from
Batch	Samples processed together: same day, kit, or run
Batch effect	A difference caused by processing, not biology
Confounded	When processing lines up with your groups, so you can't separate them
Contamination	Stray DNA from kits, reagents, or the lab
Control	A blank or known sample run alongside the real ones
Depth	How many reads each sample got
FASTQ	The text file of reads and their quality scores
Library	DNA pieces with handles and barcodes, ready for the machine
Low biomass	A sample with very little microbial DNA to start with
Metadata	Everything written down about each sample
PCR	A reaction that makes millions of copies of DNA
Quality score (Q30)	How sure the machine is; Q30 = 1 wrong letter in 1,000
Read	One piece of sequence produced by the machine
Spike-in	A known amount of something added to every sample as a ruler
UMI	A tag on each original molecule so copies are counted once
RIN	A 1-10 score of how intact RNA is

References and further reading

Everything cited today, if you want to read further.

Leek JT et al. Tackling the widespread and critical impact of batch effects in high-throughput data. Nat Rev Genet 2010.

Salter SJ et al. Reagent and laboratory contamination can critically impact sequence-based microbiome analyses. BMC Biol 2014.

Fierer N et al. Guidelines for preventing and reporting contamination in low-biomass microbiome studies. Nat Microbiol 2025.

Eisenhofer R et al. Contamination in low microbial biomass microbiome studies: issues and recommendations. Trends Microbiol 2019.

Caporaso JG et al. Global patterns of 16S rRNA diversity at a depth of millions of sequences per sample. PNAS 2011.

Caporaso JG et al. Ultra-high-throughput microbial community analysis on the Illumina HiSeq and MiSeq platforms. ISME J 2012.

Gloor GB et al. Microbiome datasets are compositional: and this is not optional. Front Microbiol 2017.

Vandeputte D et al. Quantitative microbiome profiling links gut community variation to microbial load. Nature 2017.

Silverman JD et al. Naught all zeros in sequence count data are the same. Comput Struct Biotechnol J 2020.

Bommana S et al. Metagenomic shotgun sequencing of endocervical, vaginal, and rectal samples among Fijian women. Microbiol Spectr 2022.

Ziemann M et al. Gene name errors are widespread in the scientific literature. Genome Biol 2016.

Bruford EA et al. Guidelines for human gene nomenclature. Nat Genet 2020.

EXTRA

Extra: how Oxford Nanopore works

Nanopore reads one long DNA strand as it passes through a tiny pore.



Oxford Nanopore sequencing

youtube.com/watch?v=VxGliKyYuFQ

Click to open in your browser

Watch for

1. **One strand through a tiny pore.**
   No copying needed, so reads can be very long.

2. **Small enough for the field.**
   The smallest device runs off a laptop.

3. **Software reads the current.**
   The basecaller version changes the output: record it.